H^s

THE COMPLETE MATHEMATICS

Compositional Inference on the Simplex

Peter Higgins

Rogue Wave Audio — RWA-001

April 2026

Higgins Decomposition — A deterministic 12-step pipeline for reading the geometric fingerprint of any compositional system

---

$$H^s = R \circ M \circ E \circ C \circ T \circ V \circ S$$

Contents

1. The Axiom
2. The Sample Space: Simplex Geometry
3. Aitchison Algebra
4. Log-Ratio Transforms
5. Distance and Variance
6. The Pipeline: $H^s = R \circ M \circ E \circ C \circ T \circ V \circ S$
7. Step V: Variance Trajectory and HVLD
8. Step C: The Transcendental Squeeze
9. Step E: EITT — Entropy Invariance Under Decimation
10. The O-1 Resolution: Analytic Bound on the EITT Residual
11. The EITT Decomposition Framework
12. Dimensional Collapse and the Compositional Horizon
13. Subcompositional Recursion and Amalgamation
14. Structural Modes and the Predicate Lattice
15. Theorems and Proofs
16. Canonical Constants and Benchmarks
17. Open Problems

A. Notation Reference
B. Formula Quick Reference

1. The Axiom

The entire Higgins Decomposition derives from a single statement:

$$\forall \mathcal{X} : H^s(\mathcal{X}) = H^s(\mathcal{X})$$

This says: the function is pure. Same input, same output, always. No hidden state, no randomness, no side effects. The function does not change by being called. The function does not learn from its input.

We take this axiom not as a software property but as a structural one. From this single statement, the entire function of decomposition can be derived by asking: what kind of function can satisfy this axiom when applied to compositions on the simplex?

For the axiom to hold, three structural constraints follow:

Constraint 1 — Closure. H^s maps compositions to readings. The reading does not depend on anything outside the input. Therefore H^s is a function of the composition alone.

Constraint 2 — Completeness. For every valid input, the output exists. The function is total on its domain. Therefore the domain must be bounded — there must be guards.

Constraint 3 — Invariance. The function does not change between calls. Therefore every operation inside H^s must also be deterministic. Every sub-function inherits the axiom. This is self-similarity.

Constraint 3 generates the decomposition. If the whole function is deterministic, then every part of it is deterministic. The function decomposes into parts that each satisfy the same axiom. Write H^s as a composition of sub-functions:

$$H^s = R \circ M \circ E \circ C \circ T \circ V \circ S$$

$$S = \text{simplex closure} \qquad V = \text{variance trajectory} \qquad T = \text{transform (CLR)}$$
$$C = \text{classification} \qquad E = \text{entropy test (EITT)} \qquad M = \text{mode synthesis}$$
$$R = \text{report (codes + fingerprint)}$$

Each sub-function inherits the axiom. The decomposition of H^s into sub-functions is itself a composition — parts summing to a whole. The function is compositional in the same sense that its input is compositional. The instrument has the same structure as what it measures.

2. The Sample Space: Simplex Geometry

All compositional data lives on the simplex. This chapter defines the sample space and its geometric structure.

2.1 The Open D-Part Simplex

Definition 2.1. ***The Open Simplex***

The D-part simplex is the set:

$$S^D = \{ x \in \mathbb{R}^D : x_i > 0, \ \Sigma x_i = \kappa \}$$

with $\kappa = 1$ unless stated otherwise.

Every composition $x = (x_1, x_2, \ldots, x_D)$ is a point on this manifold. The simplex is a (D−1)-dimensional surface embedded in D-dimensional Euclidean space. It is bounded (all coordinates between 0 and 1) and open (no coordinate equals zero).

2.2 The Closure Operator

Definition 2.2. ***Closure***

The closure operator projects any positive vector onto the simplex:

$$C(z) = (z_1 / \Sigma z_j, \ldots, z_D / \Sigma z_j)$$

Closure enforces the constant-sum constraint. It is deterministic, idempotent ($C(C(z)) = C(z)$), and the gateway from unconstrained data to the simplex.

2.3 The Aitchison Geometry

The simplex is not a Euclidean space. John Aitchison (1986) showed that the natural geometry of compositions requires operations that respect the ratio structure of the data.

Definition 2.3. ***Perturbation (Simplex Addition)***

$x \oplus p = C(x_1p_1, x_2p_2, \ldots, x_Dp_D)$

Perturbation is the "addition" of compositions. It is commutative, associative, and has neutral element $n = (1/D, \ldots, 1/D)$.

Definition 2.4. ***Powering (Simplex Scalar Multiplication)***

$\alpha \odot x = C(x_1^\alpha, x_2^\alpha, \ldots, x_D^\alpha)$

Powering is "scalar multiplication" on the simplex. It stretches or compresses the ratios between

components.

Together, perturbation and powering make $(S^D, \oplus, \odot)$ a real vector space. Combined with the Aitchison inner product, the simplex becomes a Euclidean vector space — a fact that is central to the entire H^s framework.

Definition 2.5. ***Aitchison Inner Product***

$\langle x, y\rangle_A = (1/D) \Sigma_{i+j} \ln(x_i/x_j) \cdot \ln(y_i/y_j)$

Equivalently: $\langle x, y\rangle_A = (1/D) \Sigma_i \mathrm{clr}(x)_i \cdot \mathrm{clr}(y)_i$

This makes $(S^D, \oplus, \odot, \langle\cdot,\cdot\rangle_A)$ a Euclidean vector space.

2.4 The Geometric Mean

The Aitchison barycenter (Fréchet mean) of a set of compositions is their geometric mean, closed to the simplex:

$$\bar{x}_G = C(\prod x_1(t)^{(1/N)}, \ldots, \prod x_D(t)^{(1/N)})$$

$$= (1/N) \odot \bigoplus x(t)$$

This is the natural centre of a set of compositions in Aitchison geometry. It is the foundation of the EITT decimation operator.

3. Aitchison Algebra

The Aitchison algebra provides the complete algebraic framework for operations on the simplex. Key identities used throughout H^s:

Perturbation Difference

$$x \ominus y = C(x_1/y_1, \ldots, x_D/y_D)$$

Measures compositional change between two states.

The Aitchison Norm

$$\|x\|_A = d_A(x, n) = \sqrt{(\langle x, x\rangle_A)}$$

where $n = (1/D, \ldots, 1/D)$ is the barycenter.

The Aitchison norm measures how far a composition departs from uniformity — its concentration. High norm means one or few components dominate; zero norm means perfect isotropy.

Key Algebraic Properties

The simplex under Aitchison operations is a (D−1)-dimensional Euclidean vector space. This means all standard linear algebra tools (projections, orthogonal decompositions, eigenanalysis) apply — but only when expressed in the correct coordinates (CLR or ILR).

4. Log-Ratio Transforms

The simplex is a constrained space. Standard statistics (means, covariances, regressions) produce spurious results on constrained data. Log-ratio transforms map compositions to unconstrained real coordinates where standard methods apply.

4.1 Centred Log-Ratio (CLR)

Definition 4.1. ***CLR Transform***

$\mathrm{clr}(x)_i = \ln(x_i) - (1/D)\,\Sigma \ln(x_j) = \ln(x_i / g(x))$

where $g(x) = (\prod x_j)^{(1/D)}$ is the geometric mean of all components.

Properties:

- Linear: $\mathrm{clr}(x \oplus y) = \mathrm{clr}(x) + \mathrm{clr}(y)$
- Zero-sum: $\Sigma\, \mathrm{clr}_i = 0$ always
- Maps S^D to a hyperplane in $\mathbb{R}^D$

The CLR is the primary coordinate system for H^s analysis. Each CLR coordinate measures how far a component is from the geometric mean of all components, in log-ratio units.

4.2 Isometric Log-Ratio (ILR)

Definition 4.2. ***ILR Transform***

$\mathrm{ilr}(x) = \Psi^T \cdot \mathrm{clr}(x) \in \mathbb{R}^{(D-1)}$

where Ψ is an orthonormal basis (Sequential Binary Partition matrix).

Each ILR balance coordinate:

$b_k = \sqrt{(r_k s_k/(r_k+s_k))} \cdot \ln(g(x^+) / g(x^-))$

where r_k, s_k are group sizes and $g(\cdot)$ is the geometric mean of the indexed parts.

The ILR is the gold standard for statistical analysis. It is a linear isometry from the simplex to $\mathbb{R}^{(D-1)}$, preserving the Aitchison inner product as the Euclidean inner product. The choice of SBP does not affect EITT — this is a theorem (Lemma L1, Chapter 15).

4.3 Pairwise Log-Ratios

$$\lambda_{ij}(t) = \ln(x_i(t) / x_j(t))$$

The pairwise log-ratio is the fundamental atom of compositional analysis. All log-ratio transforms (ALR, CLR, ILR) are linear combinations of pairwise log-ratios. The ratio-pair lattice in H^s measures the coefficient of variation of each pairwise log-ratio: locked pairs (CV < 0.01) indicate conservation laws; volatile pairs (CV > 0.50) indicate degrees of freedom.

5. Distance and Variance

5.1 Aitchison Distance

Definition 5.1. ***Aitchison Distance***

$d_A(x, y) = \|clr(x) - clr(y)\|_2$

Equivalent form:

$d_A(x, y) = \sqrt{((1/2D) \Sigma_{i+j} [\ln(x_i/x_j) - \ln(y_i/y_j)]^2)}$

Properties: metric, permutation-invariant, scale-invariant, subcompositionally coherent.

Aitchison distance is the natural metric on the simplex. It measures compositional difference in terms of log-ratio changes. It is the metric used for drift detection in H^s.

5.2 Aitchison Variance

Definition 5.2. ***Aitchison Variance (Total Variance)***

$\sigma^2_A = (1/D) \Sigma [clr_i]^2$

Equivalently: $\sigma^2_A = (1/D) \|clr(x)\|^2$

Non-negative. Zero iff x = barycenter. Measures departure from isotropy.

The Aitchison variance is the primary observable in the PLL (Phase-Locked Loop) analysis. Its trajectory over an ordered sequence of observations is the signal that H^s reads. The parabolic shape of this trajectory is the PLL lock signal.

5.3 Shannon Entropy on the Simplex

Definition 5.3. ***Shannon Entropy***

$H(x) = -\Sigma x_i \ln(x_i)$

Maximum: H = ln(D) at barycenter (1/D, ..., 1/D)

Minimum: H → 0 as x approaches a vertex

Concavity: H is strictly concave on S^D

Effective Diversity

$$N_eff = \exp(H(x))$$

The effective diversity measures how many components are "effectively present" in the composition. It is the ecological diversity equivalent. A composition with N_eff = 3.2 in a D = 7 system means that, informationally, only about 3 components carry meaningful weight.

5.4 The Hessian of Shannon Entropy

The Hessian of H on the simplex is diagonal in primal coordinates:

$$\partial^2 H / \partial x_i^2 = -1/x_i$$

This curvature is gentle in the interior (all x_i moderate) and severe at the boundary (any $x_i \to 0$). The boundary divergence is the geometric mechanism behind EITT's boundary failure mode and the compositional horizon (Chapter 12).

6. The Pipeline: $H^s = R \circ M \circ E \circ C \circ T \circ V \circ S$

The Higgins Decomposition is a deterministic 12-step pipeline with 4 input guards, 8 core steps, and an extended diagnostic panel. Every step is a pure function satisfying the axiom. The pipeline reads compositional structure without creating or destroying it.

6.1 Guard Gate (G1–G4)

The guard gate is a conjunction of four predicates. All must hold for the pipeline to proceed. Any failure is terminal.

Guard	Predicate	Failure
G1	$D \geq 2$ (at least 2 carriers)	GD-D2M-ERR → ABORT
G2	$N \geq 5$ (at least 5 observations)	GD-N5M-ERR → ABORT
G3	All values finite (no NaN/Inf)	GD-NAN-ERR → ABORT
G4	Scale disparity < 10^{15}	GD-SCL-ERR → ABORT

The guards exist because the axiom requires totality. Totality requires a bounded domain. Unbounded input can produce undefined output.

6.2 Core Steps (S4–S12)

Step 4 — Simplex Closure (S). Zero replacement (10^{-6}) followed by row normalisation to sum = 1. Projects raw data onto the simplex.

Step 5 — CLR Transform (T). Maps each observation from the simplex into CLR coordinates. Preserves ratio information. Invertible.

Step 6 — Variance Trajectory (V). Computes the cumulative Aitchison variance $\sigma^2(t)$ for $t = 1, \ldots, N$. Fits a quadratic $at^2 + bt + c$. Records the PLL R^2.

Step 7 — HVLD Vertex Lock Diagnostic. Classifies the variance trajectory: $a > 0$ → BOWL (system integrating), $a < 0$ → HILL (system segregating). Records vertex location $v = -b/(2a)$.

Step 8 — Transcendental Squeeze (C). Compares the variance trajectory against 35 transcendental constants. Classifies: NATURAL ($\delta < 0.01$), INVESTIGATE ($0.01 < \delta < 0.05$), FLAG ($\delta > 0.05$).

Step 9 — EITT Entropy Test (E). Geometric-mean block decimation at ratios M = 2, 4, 8. Tests entropy invariance. Records chaos indicators (stalls, spikes, reversals).

Steps 10–12 — Geometric Projections. Ternary projection, complex plane map, helix/polar embedding. Visualisation transforms that do not change the diagnosis.

6.3 Extended Diagnostic Panel

The extended panel comprises universal diagnostics (per-carrier contribution, match density, CoDa structural hierarchy, carrier drift, claim status) and conditional diagnostics (PID, transfer entropy, order sensitivity, ratio-pair lattice, zero crossings). Together they produce up to 78 diagnostic codes and 10 structural modes.

7. Step V: Variance Trajectory and HVLD

7.1 The Cumulative Aitchison Variance

For a compositional time series {x(t)}, the cumulative Aitchison variance at observation t is:

$$\sigma^2_A(t) = (1/D)\ \Sigma_i\ [clr_i(t)]^2$$

This trajectory is the information accumulation function — it measures how much compositional information has accumulated by observation t. The shape of this trajectory is the primary diagnostic.

7.2 The Vertex Theorem

Theorem T1. ***Vertex Theorem (Orthogonality Condition)***

Let {x(t)} be a D-part compositional time series on S^D with differentiable CLR trajectories. The vertex (extremum) of σ^2_A occurs at t = t* where:

$$d(\sigma^2_A)/dt = (2/D)\ \Sigma\ clr_i(t) \cdot clr_i'(t) = 0$$

This is equivalent to $clr(t) \perp clr'(t)$ in the Aitchison inner product.

Proof. (1) $\sigma^2_A(t) = (1/D)\ \Sigma\ [clr_i(t)]^2$ by definition. (2) Differentiate: $d(\sigma^2_A)/dt = (2/D)\ \Sigma\ clr_i(t) \cdot clr_i'(t)$. (3) Set equal to zero: $\Sigma\ clr_i(t) \cdot clr_i'(t) = 0$. (4) By definition of the Aitchison inner product: $\langle clr(t), clr'(t)\rangle = 0$, i.e., $clr(t) \perp clr'(t)$. QED.

7.3 Corollaries

Corollary C1.1: PLL Error Signal

Near the vertex t_0, $\sigma^2_A(t) \approx \sigma^2_A(t_0) + a(t - t_0)^2$ where $a = (1/D)\ \Sigma\ [(clr_i')^2 + clr_i \cdot clr_i'']$ evaluated at t_0. The error signal $\varepsilon(t) = d\sigma^2_A/dt = 2a(t - t_0)$ is linear near the lock point.

Corollary C1.2: Bowl vs Hill Classification

If a > 0 (second derivative positive), σ^2_A has a **bowl** (minimum) — the system is locked, integrating. If a < 0 (second derivative negative), σ^2_A has a **hill** (maximum) — the system is relaxing, segregating.

Corollary C1.3: Vertex Location as Transition Date

When $\sigma^2_A(t)$ is parabolic with vertex at t_0, the value t_0 marks the moment of dynamic equilibrium — where competing compositional forces are most balanced. Examples: California energy vertex 2014 (solar/wind began to dominate), Gold/Silver vertex 1751 (bimetallism equilibrium).

8. Step C: The Transcendental Squeeze

The classification must be deterministic — it must produce a discrete label, not a continuous value. A continuous trajectory must be compared to fixed reference points to produce a discrete label. The

transcendental constants satisfy the requirements: they are universal (not domain-specific), finite in number, exactly reproducible on any machine, and mathematically structured in families.

8.1 The Constant Library

The pipeline compares the variance trajectory against 35 transcendental constants drawn from families: Euler family (e, $1/e$, e^2, $\sqrt{e}$, $\ln(2)$, ...), Circle family (π, $1/\pi$, π^2, $\pi/2$, ...), Cross-family (π^e, $e\pi$, $1/(e\pi)$, ...), and Extended family (γ, φ, $\zeta(2)$, $\zeta(3)$, Khinchin K_0, Lambert W, ...).

8.2 Classification Partition

For each constant κ in the library, compute $\delta(\kappa) = \min_t |\sigma^2(t) - \kappa|$. Also test the reciprocal. Then classify:

Class	Condition	Interpretation
NATURAL	$\delta < 0.01$	Variance trajectory locks to a transcendental constant
INVESTIGATE	$0.01 \leq \delta < 0.05$	Near-match, worth examining
FLAG	$\delta \geq 0.05$	No match — may indicate missing carriers or combinatorial origin

The classification is an exhaustive partition — every composition receives exactly one label. NATURAL does not mean "good" and FLAG does not mean "bad". In the HEPData validation campaign (Hs-24), FLAG correctly identified combinatorial systems (W, Z, tau, proton) while NATURAL identified dynamical systems (Higgs, top, cosmic, CKM).

9. Step E: EITT — Entropy Invariance Under Decimation

The Entropy-Invariant Time Transformer (EITT) tests whether Shannon entropy is preserved under geometric-mean temporal decimation. It is the axiom turned inward — the function checking its own consistency under resolution change.

9.1 Block Decimation Operator

Definition 9.1. ***Block Decimation***

Let $\{x(t)\}$ be a compositional time series on S^D. For block width $M \geq 2$, define the geometric-mean block decimation operator D_M:

$$D_M[\{x(t)\}] = \{\bar{x}_G^{(b)}\} \text{ for } b = 1, \ldots, \lfloor N/M \rfloor$$

where $\bar{x}_G^{(b)} = C(\prod x_i(t)^{(1/M)})$ over each block of M consecutive compositions.

Each block of M consecutive compositions is replaced by their Aitchison-geometric mean. This is the natural averaging operation on S^D. In CLR coordinates, geometric-mean decimation becomes arithmetic averaging:

$$\mathrm{clr}(\bar{x}_G^{(b)}) = (1/M) \Sigma\, \mathrm{clr}(x(t)) \qquad \text{[exact — no approximation]}$$

This identity (Theorem T2, Chapter 15) is the algebraic heart of EITT.

9.2 The EITT Residual

Definition 9.2. ***EITT Residual***

The entropy invariance residual at decimation ratio M:

$$\Delta H_M = (1/\lfloor N/M \rfloor) \Sigma\, H(\bar{x}_G^{(b)}) - (1/N) \Sigma\, H(x(t))$$

Expressed as a percentage of base entropy:

$$\delta_M = (\Delta H_M / \bar{H}) \times 100\%$$

9.3 Empirical Observation

For stationary or slowly-varying compositional time series, the entropy residual is small: $|\delta_M| < 2\%$ when the autocorrelation boundary is satisfied. Tested across 18 domains, 44 orders of magnitude, and 5 programming languages.

Domain	D	N	M	δ_M

Gold/Silver prices (338yr)	2	339	2:1	+0.085%
World electricity	7	25	2:1	−0.589%
European prices	8	341	341:1	+0.18%
EMBER 6 countries (mean)	9	12	2:1	1.02%

9.4 Chaos Detection

EITT includes three chaos indicators measured via angular velocity ω on the ternary projection:

Stalls: $\omega(t) < 0.15 \cdot \mu_\omega$ — the trajectory nearly stops.

Spikes: $\omega(t) > 3.5 \cdot \mu_\omega$ — sudden rapid motion.

Reversals: ≥2 sign changes in angular velocity per 5 steps.

Aggregate chaos level: few deviations = smooth (S9-CHS-INF); many deviations = turbulent (S9-CHH-DIS).

9.5 EITT Inversion

When EITT fails at dimension K, adjusting K can restore invariance. **Upward inversion**: K too small → amalgamated parts hide internal structure → increase K. **Downward inversion**: K too large → individual parts are non-stationary → decrease K by amalgamation. The direction of the fix is itself diagnostic: upward identifies hidden dimensions; downward identifies non-stationary subcomponents.

9.6 Rényi/Tsallis Generalisation

The near-invariance is not a property of Shannon entropy. It is a property of the geometric-mean decimation operator acting on the simplex. Rényi entropy H_q for $q \in [0.1, 5.0]$ shows the same invariance. The mechanism: any smooth functional on the simplex whose Hessian is bounded will exhibit near-invariance when the process has small Aitchison variance.

10. The O-1 Resolution: Analytic Bound on the EITT Residual

Open Problem O-1 was the central open problem of the programme: prove a general bound on $|\delta_M|$. It was resolved on April 28, 2026.

10.1 The Theorem

Theorem T3. ***EITT Second-Order Hessian Bound***

Assume:

(A1) Interior bound: $x_i(t) \geq \delta > 0$ for all i, t.

(A2) Finite second moment: $\mathrm{Cov}(x(t)) = \Sigma_x$ with finite trace $V = \mathrm{tr}(\Sigma_x)$.

(A3) Sufficient mixing: stationary, autocorrelation decays fast enough that $\mathrm{Cov}(\bar{x}_M) = \Sigma_x/M + o(1/M)$.

Then:

$$|E[H(\bar{x}_G)] - H(x^*)| \leq (1/(2\delta M)) \cdot V + O(V^{3/2} / M^{3/2})$$

In relative form:

$$|E[\delta_M]| \leq (D-1)\, \sigma^2_A / (2\delta\, M\, \bar{H}) + O(M^{-3/2})$$

10.2 Proof Sketch

Step 1 — CLR linearity is exact. $\mathrm{clr}(\bar{x}_G) = (1/M)\, \Sigma\, \mathrm{clr}(x(t))$. This is an algebraic identity of the Aitchison barycenter. No approximation.

Step 2 — Second-order Taylor expansion. Expand H around the Fréchet mean x *using the CLR parametrisation:* $H(x(u)) = H(x) + \nabla H \cdot (u - u) + (1/2)(u - u)^T\, \mathcal{H}\, (u - u^*) + R_3$.

Step 3 — First-order cancellation. Under (A3), $E[\mathrm{clr}(\bar{x}_G)] = E[\mathrm{clr}(x(t))] = u^*$, so the linear term averages to zero. This is the exp/ln duality — the geometric mean and entropy share base e.

Step 4 — Hessian bound. The Hessian of H has entries $-1/x_i$, bounded by $1/\delta$ under (A1). Pulled through the CLR embedding: $|\mathrm{tr}(\mathcal{H} \cdot \mathrm{Cov}(u_G))| \leq (D-1)\sigma^2_A / (\delta M)$.

Step 5 — Remainder. Third-order Lagrange remainder bounded by $6/\delta^2$. Higher orders are asymptotic, not convergent.

10.3 The Closed-Form Bound (Gemini Resolution)

The temporal persistence boundary explains the 10,000x gap between empirical stability and standard VAR(1) bounds:

$$|\Delta H(M)| \leq 0.5 \cdot \lambda_{max}(|\mathcal{H}_H(\mu)|) \cdot \mathrm{Tr}(\Sigma_1) \cdot [1 - R(M, \rho)]$$

$$\text{where } R(M, \rho) = 1/M + (2/M^2) \Sigma_{k=1}^{M-1} (M-k) \rho^k$$

Three clean separation boundaries:

Condition	R(M, ρ)	Gap [1−R]	Result
ρ → 0 (synthetic noise)	1/M	≈ 1	Massive variance destruction → EITT fails
ρ → 1 (persistent system)	→ 1	→ 0	Variance conserved → EITT passes
x_i → 0 (simplex edge)	any	any	λ_max → ∞ → bound explodes

The bound mathematically necessitates the exact empirical failures observed in the adversarial test suite. Synthetic noise fails because ρ ≈ 0. Boundary compositions fail because the Hessian diverges. Persistent systems pass because variance is conserved.

11. The EITT Decomposition Framework

The decomposition framework extends EITT from a single-layer diagnostic to a three-layer instrument by treating the residual as a coordinate rather than a verdict.

11.1 The Entropy Landscape

Shannon entropy H(x) defines a scalar field on the simplex. The level sets are entropy contours — surfaces of constant entropy. At the centre of the simplex, entropy is maximal. At the boundaries, it is minimal. The Fisher-Rao metric gives this landscape Riemannian structure; the Hessian has entries $-1/x_i$.

11.2 Parallel-Perpendicular Decomposition

At any point along a compositional trajectory, the velocity decomposes into two orthogonal components:

Parallel motion v∥: Movement along entropy contours. Composition changes but entropy stays the same. EITT can safely decimate this component.

Perpendicular motion v⊥: Movement across entropy contours. Composition changes and entropy changes. This is what creates the EITT residual.

11.3 Crossing Intensity and Winding Ratio

The crossing intensity χ(t) = |v⊥(t)| / |v(t)| measures the fraction of motion directed across contours. The winding ratio W = C($\bar{H}$) / L_A measures how many times the process crosses its mean entropy contour per unit Aitchison path length. High W = oscillatory (EITT-friendly). Low W = transitional (EITT-challenging).

11.4 Three Layers

Layer 1 (Existing): The residual. Pass or fail. **Layer 2 (New):** The crossing profile. Why it passes or fails. **Layer 3 (New):** The decomposition. The entropy-preserving and entropy-changing components separated.

12. Dimensional Collapse and the Compositional Horizon

12.1 The Pruning Process

Geometric-mean decimation is a smooth, non-violent operation. It preserves robust structure and dissolves fragile structure. Components with sufficient mass remain in the simplex interior where the entropy surface is flat. Components that fall below the critical threshold enter a regime where Shannon entropy's curvature overwhelms the signal.

12.2 The Compositional Horizon

Definition 12.1. ***Compositional Horizon***

The threshold below which a component's contribution to the entropy landscape is dominated by curvature artefacts rather than genuine compositional information.

Empirical value: $\min(x_i) \approx 0.05$

Below this threshold, the Hessian entry $-1/x_i$ exceeds the signal-to-noise ratio of the EITT residual. The component is causally disconnected from the analysis.

The formal derivation of this threshold from the curvature of H on the D-simplex remains open (Open Problem D-6/O-4).

12.3 Dimensional Collapse Cycle

When a component crosses below the compositional horizon, the system undergoes dimensional collapse: the effective dimensionality drops from D to D−k. The surviving components reconstitute on a lower-dimensional simplex with higher internal coherence. The cycle is:

(1) Structure forms on a D-dimensional simplex. (2) Pruning separates weakly bound components. (3) Dimensional collapse occurs at the critical ratio. (4) Survivors reconstitute on a (D−k)-dimensional simplex with new interior stability.

The cycle is self-reinforcing: fewer components means larger shares, pushing compositions further into the interior where EITT invariance is strongest.

13. Subcompositional Recursion and Amalgamation

H^s is defined on S^D for any $D \geq 2$. Amalgamation maps $S^D \to S^{(D-1)}$ by summing carriers. The function can be applied to its own reduced output. This is recursion.

13.1 Amalgamation Operator

For index set $S \subset \{1, \ldots, D\}$ with $|S| \geq 2$, the subcomposition is $x_S = C(x_i : i \in S) \in S^{|S|}$. The pipeline generates all non-trivial amalgamation schemes $|A| = 2^D - D - 1$ and runs core diagnostics on each.

13.2 Persistence and Duality Maps

Persistence map: Which ratio locks survive all amalgamations? These are **deep structure** — conservation laws that persist regardless of how the system is partitioned.

Duality map: Locked pairs are **black holes** (information trapped). Volatile pairs are **white holes** (information escaping). Pairs hidden inside amalgamated groups are **invisible** at that resolution.

13.3 Subcompositional Coherence of EITT

If EITT holds on the full composition, it generally holds on subcompositions. Exception: subcompositions containing only non-stationary components (e.g., renewables-only in energy data) may fail. The failure is diagnostic — it identifies which subcomposition violates the stationarity assumption.

14. Structural Modes and the Predicate Lattice

Structural modes are not computed — they are deduced from combinations of atomic predicates. Ten modes emerge from approximately 30 atomic predicates.

Mode	Predicate Conjunction	Interpretation
SM-SCG-INF	BOWL ∧ SMOOTH ∧ EITT_PASS	Smooth convergence
SM-BPO-DIS	HILL ∧ TURBULENT ∧ EITT_FAIL	Bimodal/polarised
SM-RTR-DIS	EVOLVING ∧ STALLS_EXIST	Regime transition
SM-CPL-DIS	∃ LOCKED pair ∧ PID_REDUNDANCY	Carrier coupling
SM-IND-DIS	∀ pairs VOLATILE ∧ PID_UNIQUE	Independence
SM-DMR-DIS	NATURAL ∧ EULER(κ*) ∧ STRONG_FIT	Domain resonance
SM-OVC-CAL	NATURAL ∧ $R^2 > 0.99$	Overconstrained
SM-MCA-WRN	FLAG ∧ ZERO_CROSSINGS	Missing carrier alert
SM-TNT-DIS	NATURAL ∧ TURBULENT ∧ EITT_FAIL	Turbulent natural
SM-DGN-WRN	FLAG ∧ LOW_N ∧ ZERO_CROSSINGS	Degenerate

The modes are emergent properties of the predicate lattice. Each mode is a conjunction of 2–3 predicates. They represent the qualitative conclusions that follow logically from the quantitative measurements. The logic itself is compositional — modes compose from predicates as parts compose to form wholes.

15. Theorems and Proofs

15.1 Theorem T1: Vertex Theorem

Full statement and proof given in Chapter 7.

15.2 Theorem T2: CLR Linearity Identity

Theorem T2. ***CLR Linearity Identity (Exact)***

For compositions x(1), ..., x(M) on S^D, the CLR of their geometric mean equals the arithmetic mean of their CLRs:

$$\mathrm{clr}(\bar{x}_G) = (1/M)\ \Sigma\ \mathrm{clr}(x(t))$$

This is an algebraic identity. No approximation. It is the reason geometric-mean decimation preserves CLR structure exactly.

Proof. (1) $\bar{x}_G = C(\prod x(t)^{(1/M)})$ by definition. (2) $\mathrm{clr}(\bar{x}_G)_i = \ln(\bar{x}_{G,i}) - (1/D)\ \Sigma_j \ln(\bar{x}_{G,j})$. (3) $\ln(\bar{x}_{G,i}) = (1/M)\ \Sigma_t \ln(x_i(t))$ by properties of geometric mean. (4) Substituting: $\mathrm{clr}(\bar{x}_G)_i = (1/M)\ \Sigma_t\ [\ln(x_i(t)) - (1/D)\ \Sigma_j \ln(x_j(t))] = (1/M)\ \Sigma_t\ \mathrm{clr}_i(x(t))$. QED.

Significance. The arithmetic mean in CLR space IS the geometric mean on the simplex. This is the algebraic heart of EITT. It explains why geometric-mean decimation preserves structure: it is exact averaging in the natural coordinate system.

15.3 Theorem T3: EITT Hessian Bound

Full statement and proof given in Chapter 10.

15.4 Lemma L1: Balance Mean Preservation

Lemma L1. ***Balance Mean Preservation Under Decimation***

Because D_M computes the arithmetic mean in CLR space, and ILR is a linear isometry from CLR, the ILR balance coordinates of the decimated series satisfy:

$$\mathrm{ilr}_k(\bar{x}_G^{(b)}) = (1/M)\ \Sigma\ \mathrm{ilr}_k(x(t))$$

Corollary: The choice of SBP does not affect EITT. Any ILR coordinate system gives the same balance-mean values under decimation.

Proof. Follows from T2 (CLR linearity) and the linearity of the ILR transform (Ψ is a constant orthonormal matrix). QED.

15.5 Lemma L2: Shannon Entropy Concavity

Lemma L2. ***Shannon Entropy Concavity***

H(x) is strictly concave on S^D. By Jensen's inequality: H(E[X]) ≥ E[H(X)].

For block decimation: H($\bar{x}$_G) ≥ (1/M) Σ H(x(t)).

This creates a systematic upward bias in entropy under decimation. The EITT residual measures how small this bias actually is.

15.6 Compositional Uncertainty Principle

Principle. ***Compositional Uncertainty Principle (Higgins)***

Let x(t) be an oscillating D-part composition with at least one carrier approaching zero ($x_i \to 0$). Then the EITT angular velocity deviates at every zero-crossing.

The deviation floor scales as: f(N) = 6.37 × N^0.18

The exponent α = +0.18 ± 0.02 is independent of system physics, integration method, timestep, or initial conditions.

Evidence: EXP-17, 35 simulations, CV < 1.2%.

The floor is topological because the simplex boundary singularity exists in every dimension D ≥ 2. At a carrier zero-crossing, $clr_i \to -\infty$, creating a curvature spike in the Fisher information metric.

15.7 Force-Matter Asymmetry

Theorem. ***Force-Matter Asymmetry (Higgins)***

For any physical system decomposed simultaneously into force carriers and matter/energy carriers, the force simplex exhibits strictly more EITT deviations than the matter simplex.

Forces are reversible and turbulent; matter is irreversible and directional.

Evidence: EXP-18, 41 vs 5 deviations (8.2:1). EXP-18b Bridge 5: 296 vs 0.

16. Canonical Constants and Benchmarks

16.1 EITT Benchmarks

Domain	M	δ_M	Note
European prices	341:1	0.18%	Highest compression tested
EMBER 6 countries (mean)	monthly	1.02%	Energy grids
NGFS geometric 5yr	5:1	1.8%	Climate scenarios
NGFS arithmetic 5yr	5:1	14.2%	WRONG mean — 7.9× destruction
Gold/Silver 338yr	50:1	0.60%	Longest time series
QGP STAR BES-I + ALICE	—	0.27%	Highest energy
Geochemistry	—	37/39 TAS	Largest sample (40,666 rocks)

16.2 PLL Benchmarks

Metric	Value	Domain
Domains tested	30+	—
Confidence threshold exceeded by	$3^3 = 27$	All domains
Highest R^2	0.998	Room acoustics
Gold/Silver R^2	0.9007	Commodities
Gold/Silver vertex	1751	Bimetallism equilibrium
Gold/Silver Q-factor	22.18	Commodities

16.3 Operating Envelope

Threshold	Value	Interpretation
EITT pass	< 1%	Below: PASS. Above: investigate
Entropy minimum	H > 1.0	Below: boundary regime
Zero rate maximum	< 15%	Above: replacement artefacts
Zero rate hard fail	> 35%	EITT correctly fails
Compositional horizon	$\min(x_i) \approx 0.05$	Below: dimensional collapse regime

17. Open Problems

The following problems remain formally open as of April 2026. O-1 was resolved on April 28, 2026.

ID	Status	Statement
O-1	RESOLVED	General bound on $\|\delta_M\|$. Closed-form: $\|\Delta H(M)\| \le 0.5 \cdot \lambda_max(\|\mathcal{H}_H(\mu)\|) \cdot Tr(\Sigma_1) \cdot [1 - R(M, \rho)]$.
O-2	OPEN	PLL parabola genericity: prove $\sigma^2_A(t)$ is generically parabolic under mild regularity.
O-3	OPEN	Noise conjugacy: prove lock noise + anti-lock noise = 0.
O-4	OPEN	Compositional horizon formalisation: derive $\min(x_i) \approx 0.05$ from curvature of H on the D-simplex.
O-5	OPEN	Rényi invariance: prove geometric-mean decimation preserves the Rényi entropy family.
O-6	OPEN	Shape functional invariance: prove any function of ratios is EITT-invariant in the interior.
O-7	OPEN	Tail bound: concentration inequality for $\|\delta_M - E[\delta_M]\|$ using mixing coefficients.
O-8	OPEN	Non-stationary extension: extend EITT bounds to slowly-drifting processes.
D-2	OPEN	Winding ratio critical values as a function of D and mean composition.
D-6	OPEN	Compositional horizon formalisation (connected to O-4).
D-7	OPEN	Dimensional collapse dynamics: characterise pruning across successive decimation scales.

Appendix A: Notation Reference

Symbol	Definition	Note
S^D	D-part simplex: $\{x : x_i > 0, \Sigma x_i = 1\}$	Sample space
$C(\cdot)$	Closure: $C(z) = z / \Sigma z_i$	Projects onto simplex
$\oplus$	Perturbation: $x \oplus p = C(x_i p_i)$	Simplex addition
$\odot$	Powering: $\alpha \odot x = C(x_i^\alpha)$	Simplex scalar multiplication
clr	$\mathrm{clr}(x)_i = \ln(x_i) - (1/D)\Sigma \ln(x_j)$	Centred log-ratio
ilr	$\mathrm{ilr}(x) = \Psi^T \cdot \mathrm{clr}(x)$	Isometric log-ratio
d_A	$d_A(x,y) = \lVert \mathrm{clr}(x) - \mathrm{clr}(y) \rVert_2$	Aitchison distance
σ^2_A	$(1/D)\ \Sigma\ \mathrm{clr}_i^2$	Aitchison variance
H(x)	$-\Sigma\ x_i \ln(x_i)$	Shannon entropy
D_M	Block decimation operator	Geometric-mean blocks
δ_M	$\lvert \Delta H_M / \bar{H} \rvert \times 100\%$	EITT residual (%)
N_eff	$\exp(H(x))$	Effective diversity
χ(t)	$\lvert v\perp(t) \rvert / \lvert v(t) \rvert$	Crossing intensity
W	$C(\bar{H}) / L_A$	Winding ratio

Appendix B: Formula Quick Reference

The complete mathematical vocabulary of H^s in eleven formulas:

1. Closure

$$C(z) = z / \Sigma z_i$$

2. Perturbation

$$x \oplus p = C(x_1 p_1, \ldots, x_D p_D)$$

3. CLR Transform

$$\mathrm{clr}(x)_i = \ln(x_i) - (1/D)\, \Sigma \ln(x_j)$$

4. Aitchison Distance

$$d_A(x, y) = \|\mathrm{clr}(x) - \mathrm{clr}(y)\|_2$$

5. Balance

$$b = \sqrt{rs/(r+s)} \cdot \ln(g(x^+) / g(x^-))$$

6. Perturbation Velocity

$$v(t) = d_A(x(t), x(t-1))$$

7. Effective Diversity

$$N_{eff} = \exp(-\Sigma\, x_i \ln(x_i))$$

8. EITT Residual

$$\delta_M = |\Delta H_M / \bar{H}| \times 100\%$$

9. Analytic Bound (O-1)

$$|\Delta H(M)| \leq 0.5 \cdot \lambda_{max}(|\mathcal{H}_H(\mu)|) \cdot \mathrm{Tr}(\Sigma_1) \cdot [1 - R(M, \rho)]$$

10. Variance Trajectory Vertex

$$t^* \text{ where } \Sigma\, \mathrm{clr}_i(t^*) \cdot \mathrm{clr}_i'(t^*) = 0 \qquad (\mathrm{clr} \perp \mathrm{clr}')$$

11. Compositional Floor

$$f(N) = 6.37 \times N^{0.18}$$

$$H^s = R \circ M \circ E \circ C \circ T \circ V \circ S$$

The function that reads the geometric fingerprint of any compositional system.

Peter Higgins — Rogue Wave Audio — April 2026